

Computer Organization And Assembly Language

```
00000000: 01001101 01011010 10010000 00000000 00000011 00000000 MZ....
00000006: 00000000 00000000 00000100 00000000 00000000 00000000 .....
0000000c: 11111111 11111111 00000000 00000000 10111000 00000000 .....
00000012: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000018: 01000000 00000000 00000000 00000000 00000000 00000000 @.....
0000001e: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000024: 00000000 00000000 00000000 00000000 00000000 00000000 .....
0000002a: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000030: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000036: 00000000 00000000 00000000 00000000 00000000 00000000 .....
0000003c: 10000000 00000000 00000000 00000000 00001110 00011111 .....
00000042: 10111010 00001110 00000000 10110100 00001001 11001101 .....
00000048: 00100001 10111000 00000001 01001100 11001101 00100001 !..L.!
0000004e: 01010100 01101000 01101001 01110011 00100000 01110000 This p
00000054: 01110010 01101111 01100111 01110010 01100001 01101101 rogram
0000005a: 00100000 01100011 01100001 01101110 01101110 01101111 canno
00000060: 01110100 00100000 01100010 01100101 00100000 01110010 t be r
00000066: 01110101 01101110 00100000 01101001 01101110 00100000 un in
0000006c: 01000100 01001111 01010011 00100000 01101101 01101111 DOS mo
00000072: 01100100 01100101 00101110 00001101 00001101 00001010 de...
00000078: 00100100 00000000 00000000 00000000 00000000 00000000 $. ....
0000007e: 00000000 00000000 01010000 01000101 00000000 00000000 ..PE..
```

Lab Manual 09

Lab Instructor:

Mr. Tariq Mehmood Butt

tariq.butt@pucit.edu.pk

Teacher Assistants:

Hafiz Muhammad Ahmad

bcsf21m502@pucit.edu.pk

Syed Muhammad Zain Raza

bcsf21m510@pucit.edu.pk

Zahra Malik

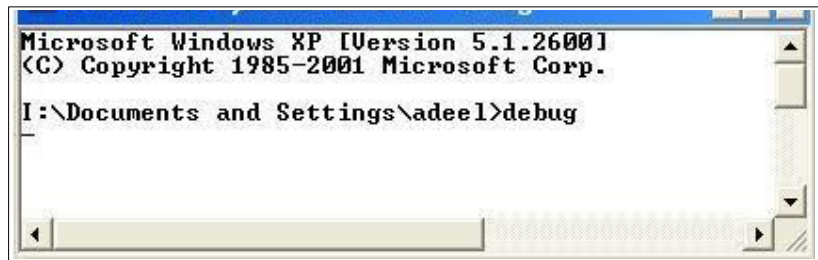
bcsf21m551@pucit.edu.pk

Bilal

bsdsf21m022@pucit.edu.pk

DEBUG provides **various** commands.

To invoke any of these commands, we enter the command character in either upper or lower case, and then enter any parameter that the command may require. Commands are executed when you press enter, not as you type them. You can change the line by pressing the backspace key to delete what you have typed and then type the correct characters. When the command line is as you want it, then press enter to execute the entire line.



Debug does not detect errors until you have pressed Enter. If you make a syntax error in an entry, the command line is displayed with the word error added at the point at which the error was detected. If a command line contains more than one error, only the first error is detected and indicated, and execution ceases at that point. Some of the commands are as following.

1. Display Help Screen (?)
2. Assemble (A)
3. Compare (C)
4. Dump (D)
5. Enter (E)
6. Fill (F)
7. GO (G)
8. Hexadecimal Arithmetic (H)
9. Input (I)
10. Load (L)
11. Move (M).
12. Name (N)
13. Output (O)
14. Proceed (P), Trace (T)
15. Quit (Q)
16. Register (R)
17. Search (S)
18. Unassemble (U)

One of the most important commands is DUMP command.

Dump (D)

The Dump command is one we use often. It displays the contents of a series of memory locations. The syntax for this command is

-D *address1 address2*

We must specify *address1*, an optional starting address for the display, before you can specify *address2*, an optional ending address. If no addresses are specified, DEBUG starts displaying memory locations with DS:0100 or, (if Dump already has been used) with the byte following the last byte displayed by the most recent Dump command.

Dump always displays 16 bytes per line, beginning with the nearest paragraph (16-byte) boundary. This display rule may differ with the first and last lines displayed, because you may have asked DEBUG to start the dump with a memory location that was not on a paragraph boundary. If you do not specify an ending address, DEBUG always displays 128 bytes of memory. Each byte is shown in both Hexadecimal and ASCII representation.

```
I:\DOCUMENT1\adeel>debug
-
-d
0B33:0100 B8 34 00 BB 43 00 01 D8-99 80 3E 20 99 00 75 24 .4..C.....> ..u$
0B33:0110 A2 24 99 0A C9 75 1D 0A-C0 74 19 8B 34 00 22 0B .$. .u...t..4."
0B33:0120 13 B0 1A 06 33 FF 8E 06-00 96 F2 AE 07 75 05 4F ....3.....u.0
0B33:0130 89 3E 21 96 BB 06 97 80-3E 13 96 00 74 03 BB 4C .>!.>.....>...t..L
0B33:0140 97 BE C3 96 8B 3E 05 98-B9 08 00 AC 3C 3F 75 02 .....>.....<?u.
0B33:0150 8A 07 3C 20 74 01 AA 43-E2 F1 B1 03 B0 20 38 04 ..< t..C..... 8.
0B33:0160 74 12 B0 2E AA AC 3C 3F-75 02 8A 07 3C 20 74 01 t.....<?u...< t.
0B33:0170 AA 43 E2 F1 32 C0 AA C3-F6 46 04 02 75 43 8B D5 .C..2....F..uC..
-
-d 0010 001F
0B33:0010 97 05 17 03 97 05 5F 04-01 01 01 00 02 FF FF FF ....._.....
-
-d 0010 L 10
0B33:0010 97 05 17 03 97 05 5F 04-01 01 01 00 02 FF FF FF ....._.....
-
```

Lab Task

Unfortunately, some of your classmates' dump instruction is not working. You are supposed to help them by implementing such a program which works like dump instruction.

Implement a program which takes single address as input from user and displays 128 byte data from that address as dump instruction.

Note:

- The user can enter address upto four characters, if less characters are entered, append remaining zeros. i.e. if user enters 2A, convert it to 002A.
- Take base address from data segment address.